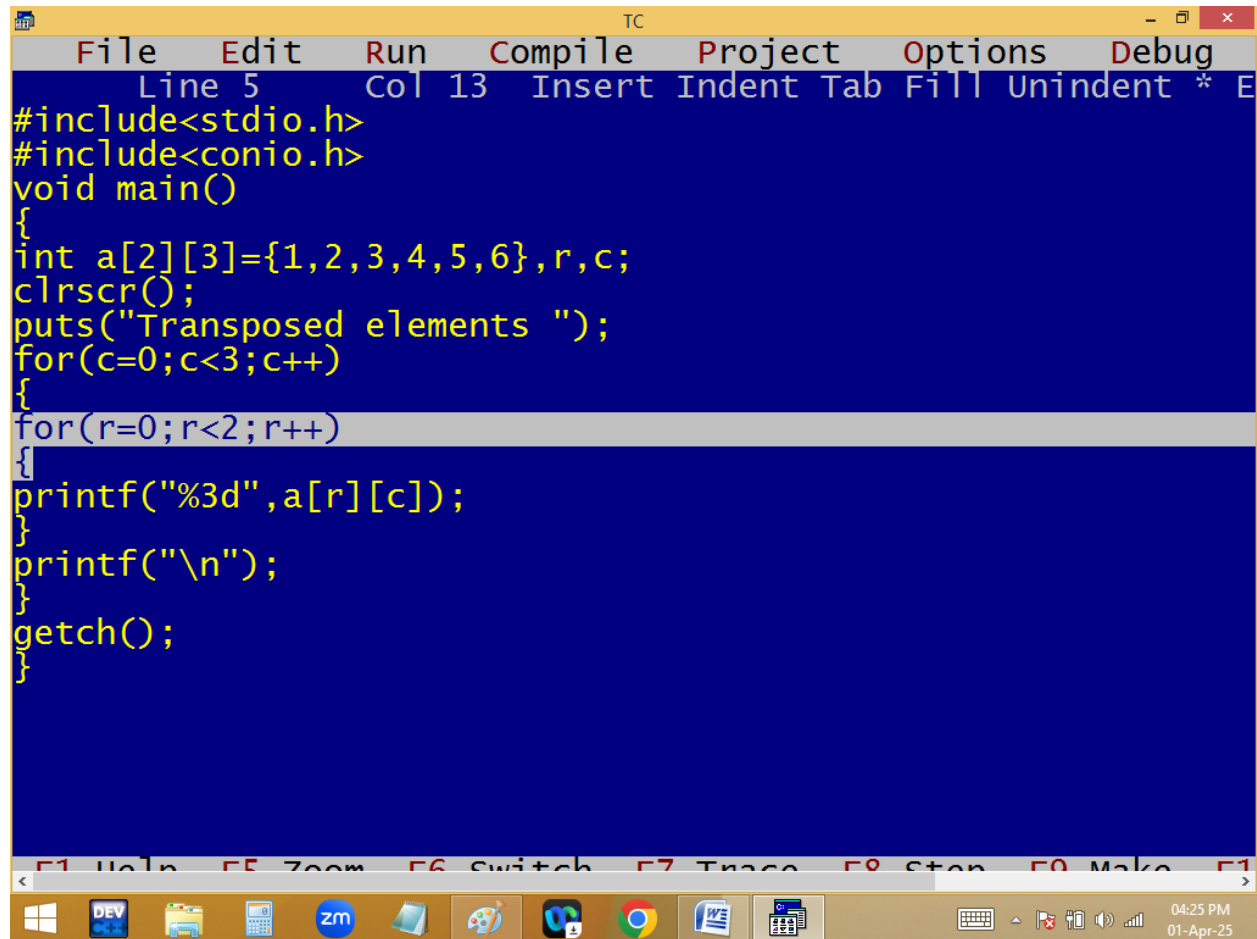


Transpose of n*n matrix:



The image shows a screenshot of the Turbo C++ (TC) IDE. The main window displays a C program designed to transpose a 2x3 matrix. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a[2][3]={1,2,3,4,5,6},r,c;
    clrscr();
    puts("Transposed elements ");
    for(c=0;c<3;c++)
    {
        for(r=0;r<2;r++)
        {
            printf("%3d",a[r][c]);
        }
        printf("\n");
    }
    getch();
}
```

The IDE's menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the bottom indicates the current cursor position as Line 5, Col 13. The Windows taskbar at the very bottom shows various application icons and the system clock set to 04:25 PM on 01-Apr-25.

```

TC
Transposed elements
1 4
2 5
3 6

```

```

for( c=0; c<3;c++)
{
for( r=0; r<2; r++ )
{
p( a[ r ][ c ] );
}
p("\n"); ✓
}

```

$\begin{matrix} r & c \\ 0 & 1 \end{matrix}$ ~~2~~ 0
 $\begin{matrix} 0 & 1 \end{matrix}$ ~~2~~ 1
 $\begin{matrix} 0 & 1 \end{matrix}$ ~~2~~ 2
~~3~~

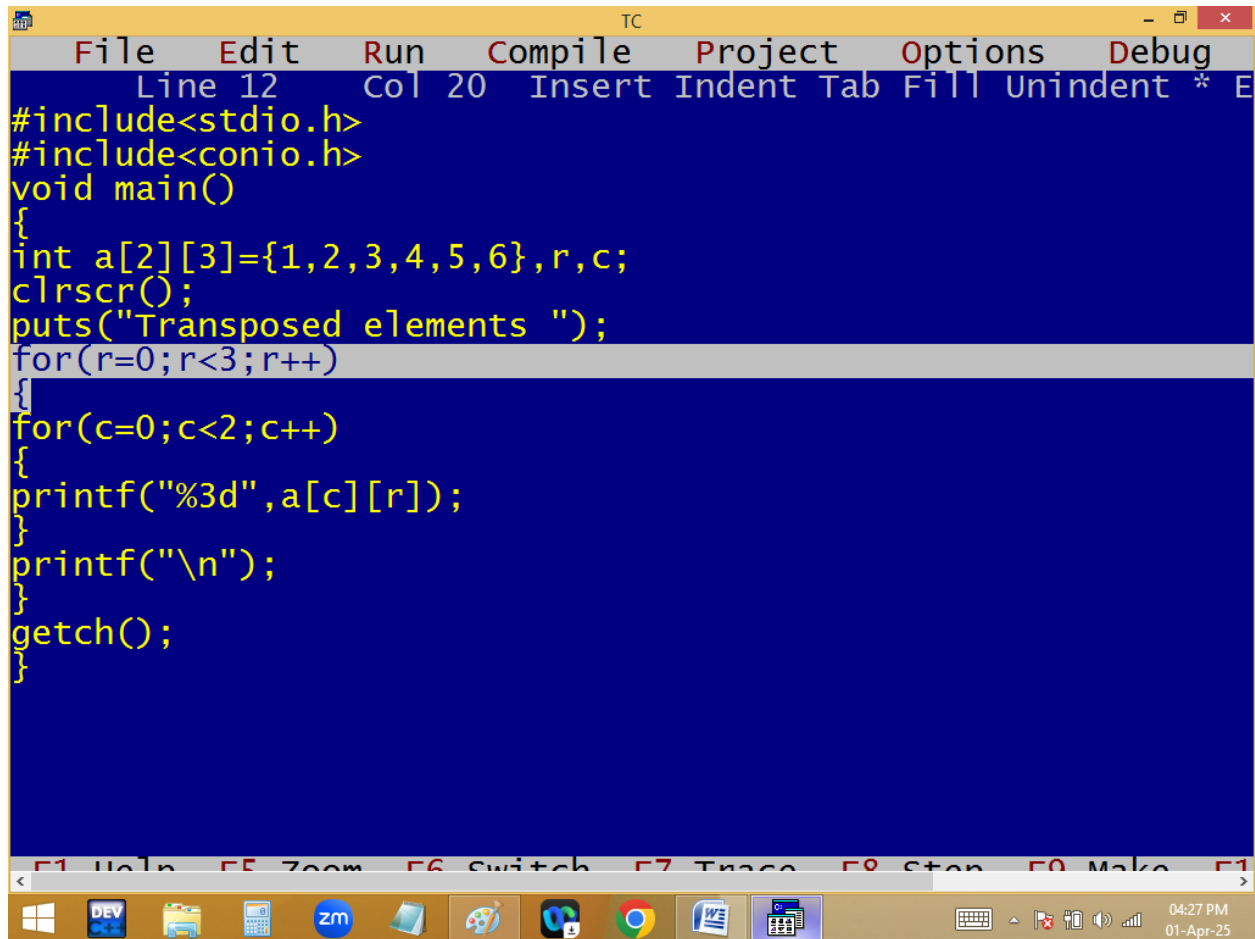
2*3

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2

3*2

✓ 1	✓ 4
✓ 2	✓ 5
✓ 3	✓ 6

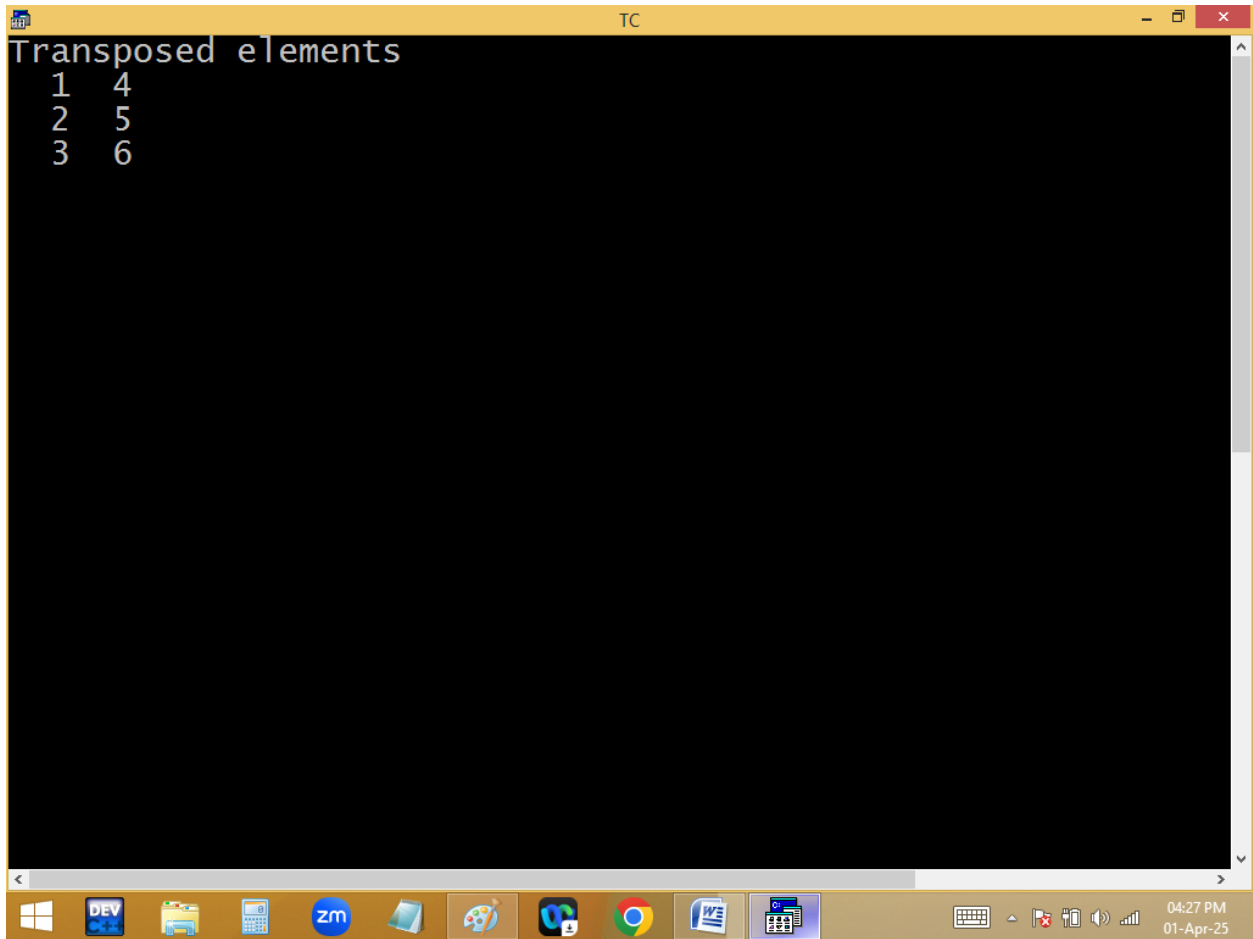
Method2:



The image shows a screenshot of the Turbo C++ (TC) IDE. The main window displays a C program for transposing a 2x3 matrix. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
puts("Transposed elements ");
for(r=0;r<3;r++)
{
for(c=0;c<2;c++)
{
printf("%3d",a[c][r]);
}
printf("\n");
}
getch();
}
```

The IDE's menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the bottom shows the time as 04:27 PM on 01-Apr-25. The Windows taskbar at the very bottom contains icons for various applications including DEV, File Explorer, Calculator, Zoom, and others.



2*3

3*2

```
for( r=0; r<3; r++ )
{
  for( c=0; c<2; c++ )
  {
    p( a[c][ r ] );
  }
  p("\n");
}
```

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2

✓1	✓4
2✓	5✓
3✓	6✓

c r
 0 1 ~~2~~ 0
 0 1 ~~2~~ 1
 0 1 ~~2~~ 2
 ~~3~~

```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 34 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Transposed elements ");
for(r=0;r<nc;r++)
{
for(c=0;c<nr;c++)
{
printf("%3d",a[c][r]);
}
printf("\n");
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:32 PM
01-Apr-25

```
TC
Enter no of rows and columns 5 2
Enter 10 elements
1 2
3 4
5 6
7 8
9 0
Transposed elements
1 3 5 7 9
2 4 6 8 0
```

```
TC
Enter no of rows and columns 3 3
Enter 9 elements
1 2 3
4 5 6
7 8 9
Transposed elements
  1  4  7
  2  5  8
  3  6  9
```

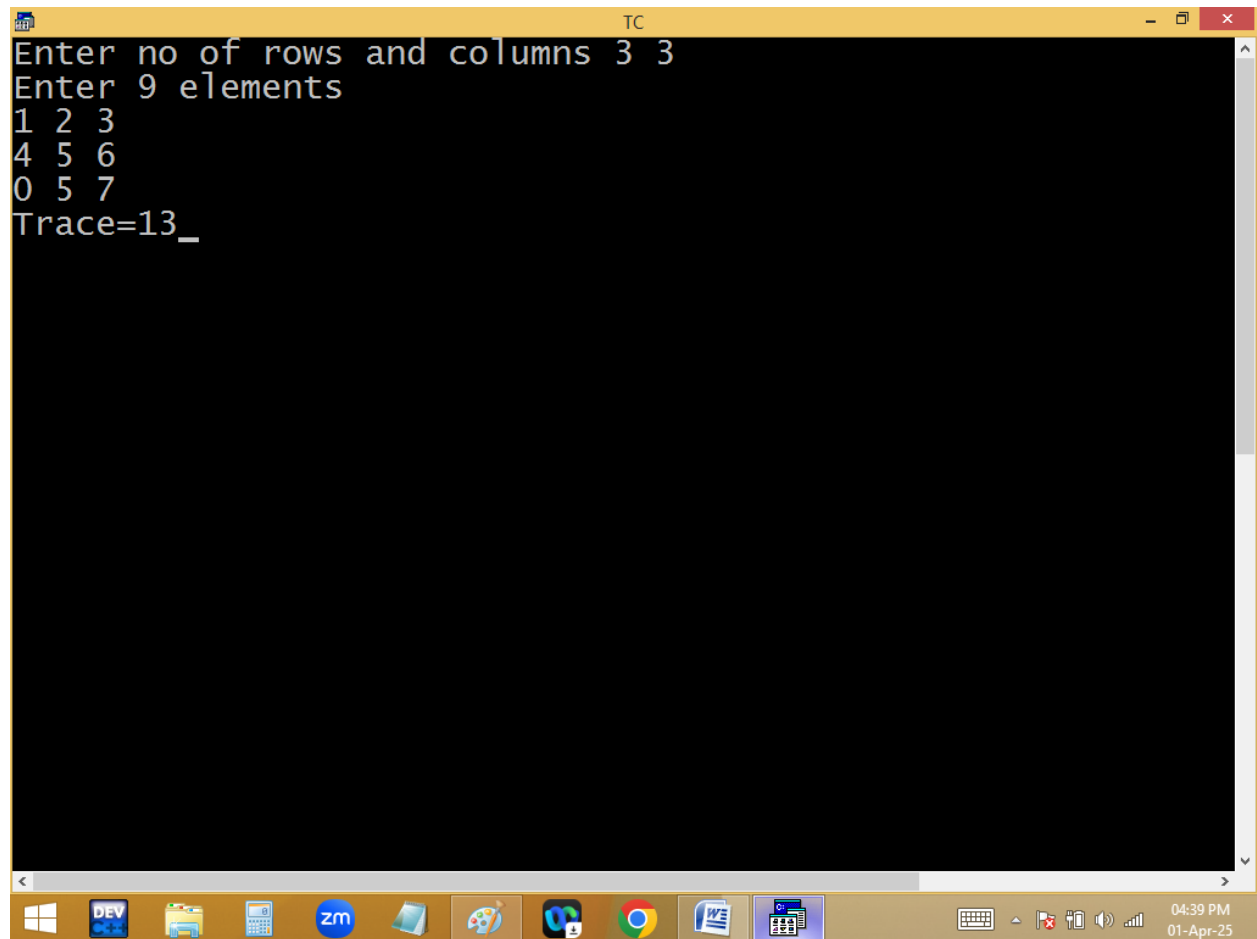
Finding trace of $n \times n$ matrix:

```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 9 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,s=0;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
if(nr==nc)
{
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)
{scanf("%d",&a[r][c]); if(r==c)s+=a[r][c];}
printf("Trace=%d",s);
}
else printf("Rows and columns should be same");
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:40 PM
01-Apr-25


```
TC
Enter no of rows and columns 3 3
Enter 9 elements
1 2 3
4 5 6
0 5 7
Trace=13_
```

A screenshot of a Windows 10 desktop environment. At the top, a terminal window titled 'TC' is open, displaying the output of a C program. The program prompts for the number of rows and columns (3 3) and 9 elements, which are entered as 1 2 3, 4 5 6, and 0 5 7. The final output is 'Trace=13_'. The taskbar at the bottom shows various application icons including Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. System tray icons on the right indicate the time as 04:39 PM on 01-Apr-25.

```
TC
Enter no of rows and columns 2 3
Rows and columns should be same
```

Sum of principle diagonal elements

if($r==c$) $s+=a[r][c]$;

<u>r</u>	<u>c</u>	<u>s</u>
0	0 1 2	$0+1+5+7=13$
1	0 1 2	
2	0 1 2	

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2
0 2,0	5 2,1	7 2,2

Trace = 13

Finding trace of right diagonal elements:

```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 36 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,s=0;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
if(nr==nc)
{
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)
{scanf("%d",&a[r][c]); if(r+c==nr-1)s+=a[r][c];}
printf("Trace=%d",s);
}
else printf("Rows and columns should be same");
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:43 PM
01-Apr-25

```
TC
Enter no of rows and columns 3 3
Enter 9 elements
1 2 0
3 4 5
9 8 4
Trace=13
```

Sum of right diagonal elements: 8

if(r+c == nr-1) s+=a[r][c];

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2
0 2,0	5 2,1	7 2,2

nr
 $3-1=2$

```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\tEven\tOdd\tZero");
puts("-----");
for(r=0;r<nr;r++)
{
for(c=e=o=z=0;c<nc;c++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-row\t%d\t%d\t%d\n",r+1,e,o,z);
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:04 PM
01-Apr-25

```
TC
Enter no of rows and columns 3 3
Enter 9 elements
1 2 3
4 5 6
0 5 7

Even      Odd      Zero
-----
1-row     1        2        0
2-row     2        1        0
3-row     0        2        1
```

Windows taskbar: DEV, File Explorer, Calculator, ZOOM, Paint, VS Code, Chrome, Word, Excel, System tray (05:05 PM, 01-Apr-25)

```

TC
Enter no of rows and columns 4 5
Enter 20 elements
1 2 3 4 0
4 6 8 2 0
1 3 5 7 0
1 3 2 4 5
      Even      Odd      Zero
-----
1-row  2         2         1
2-row  4         0         1
3-row  0         4         1
4-row  2         3         0

```

`p("\t Even\tOdd\tZero");` ✓
`p("-----");`

```

for(r=0;r<nr;r++)
{
for(e=o=z=c=0;c<nc;c++)
{
if(a[r][c]==0)z++;
else if(a[r][c]%2==0)e++;
else o++; ✓
}
p("%d-row\t%d\t%d\t%d\n",r+1,e,o,z);
}

```

1 0,0	2 0,1	3 0,2
4 1,0	5 1,1	6 1,2
0 2,0	5 2,1	7 2,2

	Even	Odd	Zero
1-row	1	2	0
2-row	2	1	0
3-row	0	2	1

$\begin{matrix} r & c & e & o & z \\ 0 & 0 & 1 & 2 & 0 \\ 1 & 0 & 1 & 2 & 0 \\ 2 & 0 & 1 & 2 & 0 \\ & 1 & 0 & 1 & 2 \\ & 1 & 0 & 1 & 2 \\ & 2 & 0 & 1 & 2 \end{matrix}$

Column wise:


```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 8 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\tEven\tOdd\tZero");
puts("-----");
for(c=0;c<nc;c++)
{
for(e=o=z=r=0;r<nr;r++)
{
if(a[r][c]==0)z++;else if(a[r][c]%2==0)e++;else o++;
}
printf("%d-col\t%d\t%d\t%d\n",c+1,e,o,z);
}
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:17 PM
01-Apr-25

```
TC
Enter no of rows and columns 3 4
Enter 12 elements
1 2 3 0
2 4 6 1
1 3 5 7
Even      Odd      Zero
-----
1-col     1        2        0
2-col     2        1        0
3-col     1        2        0
4-col     0        2        1
```

Finding row sum and columns sum:

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void main()
```

```
{
```

```
int a[10][10]={0},nr,nc,r,c,rs,cs;
```

```
clrscr();

printf("Enter no of rows and columns ");

scanf("%d%d",&nr,&nc);

printf("Enter %d elements ",nr*nc);

for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][
c]);

for(r=0;r<nr;r++)
{
for(c=rs=cs=0;c<nc;c++)
{
rs+=a[r][c]; cs+=a[c][r];
}

a[r][c]=rs; a[c][r]=cs;
}

puts("Elements are ");

for(r=0;r<=nr;r++)
```

```
{  
for(c=0;c<=nc;c++)  
{  
    if(r==nr && c==nc)continue; else  
    printf("%4d",a[r][c]);  
}  
printf("\n");  
}  
getch();  
}
```

```
TC
Enter no of rows and columns 2 2
Enter 4 elements
1 6
9 5
Elements are
  1  6  7
  9  5 14
 10 11
```

Activate Windows
Go to PC settings to activate Windows.

Home work?

Finding the fractions of $n \times n$ matrix:

a		b			
9.9	4.4	6.6	7.2	1.5	0.61
0,0	0,1	0,0	0,1		
7.7	1.1	5.5	1.01	1.4	1.08
1,0	1,1	1,0	1,1		

Fractions of $n \times n$ matrix

Matrix Multiplication?

